

TableMate

Auditable bimanual manipulation in simulation

A reproducible MuJoCo workstation for two SO-101 arms

THE PROBLEM

A dinner instruction becomes inspectable action

Robotic table setting fails silently when perception, contact, and coordination are hidden.

TableMate keeps each decision and physical consequence visible: camera observations enter a planner, a supervisor selects a bounded skill, and MuJoCo verifies contacts and limits.

Operator needs

Fresh camera state

Task progress and stop control

Replayable evidence after every attempt

ARCHITECTURE

A layered system with clear ownership

Instruction +
cameras

Vision-language planner

Task supervisor

ACT skill policy

Contact and action checks

MuJoCo: two SO-101 arms, drawer, utensils, plate, cup

The supervisor owns prerequisites, arm assignment, timeouts, cancellation, recovery, and the final success decision.

Physics is part of the evidence

Contact-based manipulation

Grasps and hand-offs use MuJoCo contacts. Objects are never teleported or welded to a gripper.

Reproducible operations

Versioned contracts, run manifests, dataset lineage, recovery limits, CLI workflows, FastAPI portal, and seven lessons.

Bounded learned actions

LeRobot ACT predicts joint targets from camera images and robot observations. Invalid, stale, and out-of-range actions are rejected.

Native execution

The local development path uses verified ARM64 Python and keeps CPU and MPS evidence separate.

MEASURED RESULT

The system stopped safely when the policy was unsafe

Teacher foundation

A complete contact-based teacher layout passed with synchronized camera replay and physical checks.

Learned evaluation

The CPU ACT candidate completed 20,000 updates. During physical placement, the contact guard stopped at 2.537 mm workbench overlap after 381 autonomous actions.

Interpretation

The run is recorded as a failure. The guard did not silently claim completion. Five corrective teacher trajectories then passed and remain separately labeled as teacher data.

NEXT STEPS

Reproduce the evidence, then extend the envelope

Run locally

scripts/bootstrap.sh
scripts/check.sh --render
open the portal and replay retained runs

Complete the release gate

Finish sealed export parity, retrain from a fresh declaration, evaluate held-out scenes, and execute the final benchmark on real Intel Core Ultra hardware.

Repository

github.com/lipengyuan1994/bimanual-robotic-manipulation

Public lessons: lipengyuan1994.github.io/bimanual-robotic-manipulation/